only for digital currency such as that offered by Zcash, a similar solution can be adapted to other application domains, such as health care, financial, and military.

12.9 Security

Security is a challenge for any computing system and network, especially in an open and decentralized system of typically unknown participants. Over the years, security at the network level (http:// to https://), infrastructure level (firewalls), system level (dual authentication passwords), and similar measures have improved the overall security of networked systems. Typically, a blockchain-based application is part of such a system. In addition, robust cryptographic algorithms and hashing algorithms have helped secure blockchain at the protocol level and the application level. Here are some of the approaches used:

- *256-bit processors and computations for blockchain operations* --The 256-bit address space is four times exponentially larger than 64-bit space, and a larger address space means that the probability of hashing collision is lower, thus preserving the integrity and security of blockchain operation.
- *The private-public key pair* --The private-public key pair is the metaphorical passport to participating and transacting on the blockchain. Similar to the way you use, secure, and protect a credit card, you need to protect the private key and the mnemonic representing it for the security of your assets.